

Building a Power Supply

Project lab report

ETE 2721L.01 CAD Mnfctrng PCB Fbrctn Lab

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**Cal Poly
Pomona**

College of Engineering

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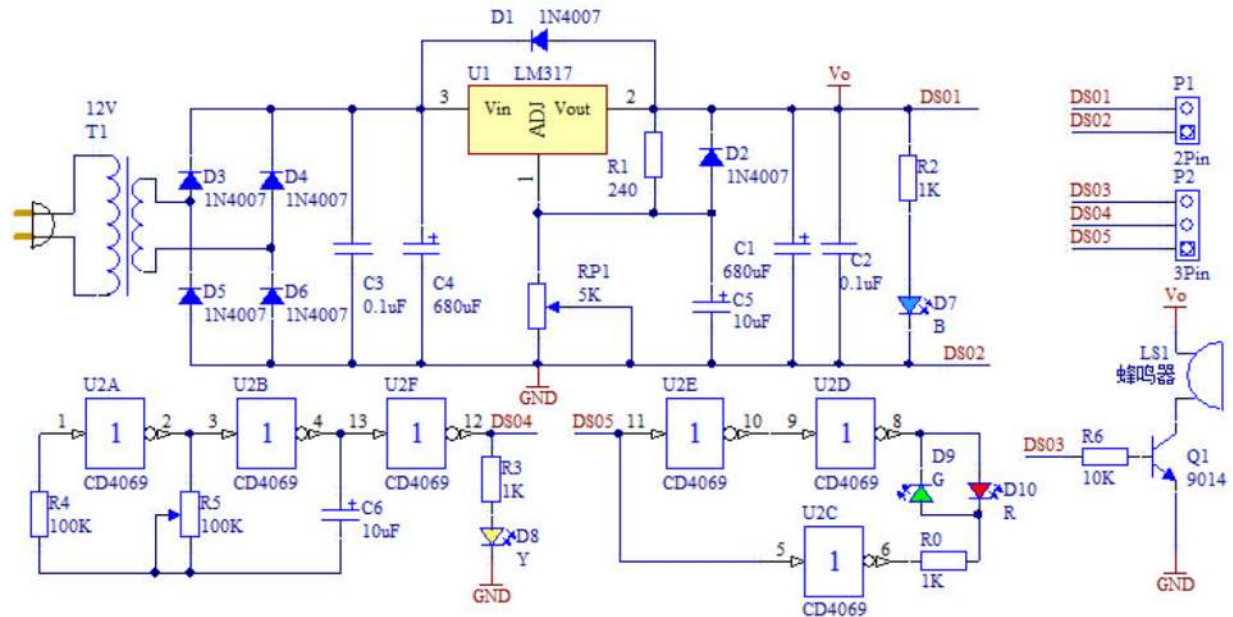
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Introduction

The goal of this project is to convert AC voltage (120v from a wall outlet) into an adjustable DC voltage output. One PCB was designed for this project (using Altium) , but three separate boards were soldered and tested. Two of these attempts used the PCB designed by us (Edwin and Sarah), while the third attempt used a board from Alinan US. The PCB designed by Alinan US was used as an attempt with the goal of it being a backup. This PCB served as a safety net because it's functioning was more certain because it was designed by a cooperation rather than by students who were beginners with using Altium. The decision to solder and work with a backup board in addition to our designed PCB arose due to concerns about the PCB design, clearances and issues with soldering. Conveniently, the backup PCB uses an almost identical circuit and the same components, with the same functioning goals. More information about the PCB used for the third attempt can be found in the links section of the report. Information about the overall results of this project can be found in the soldering and project results section of the report.

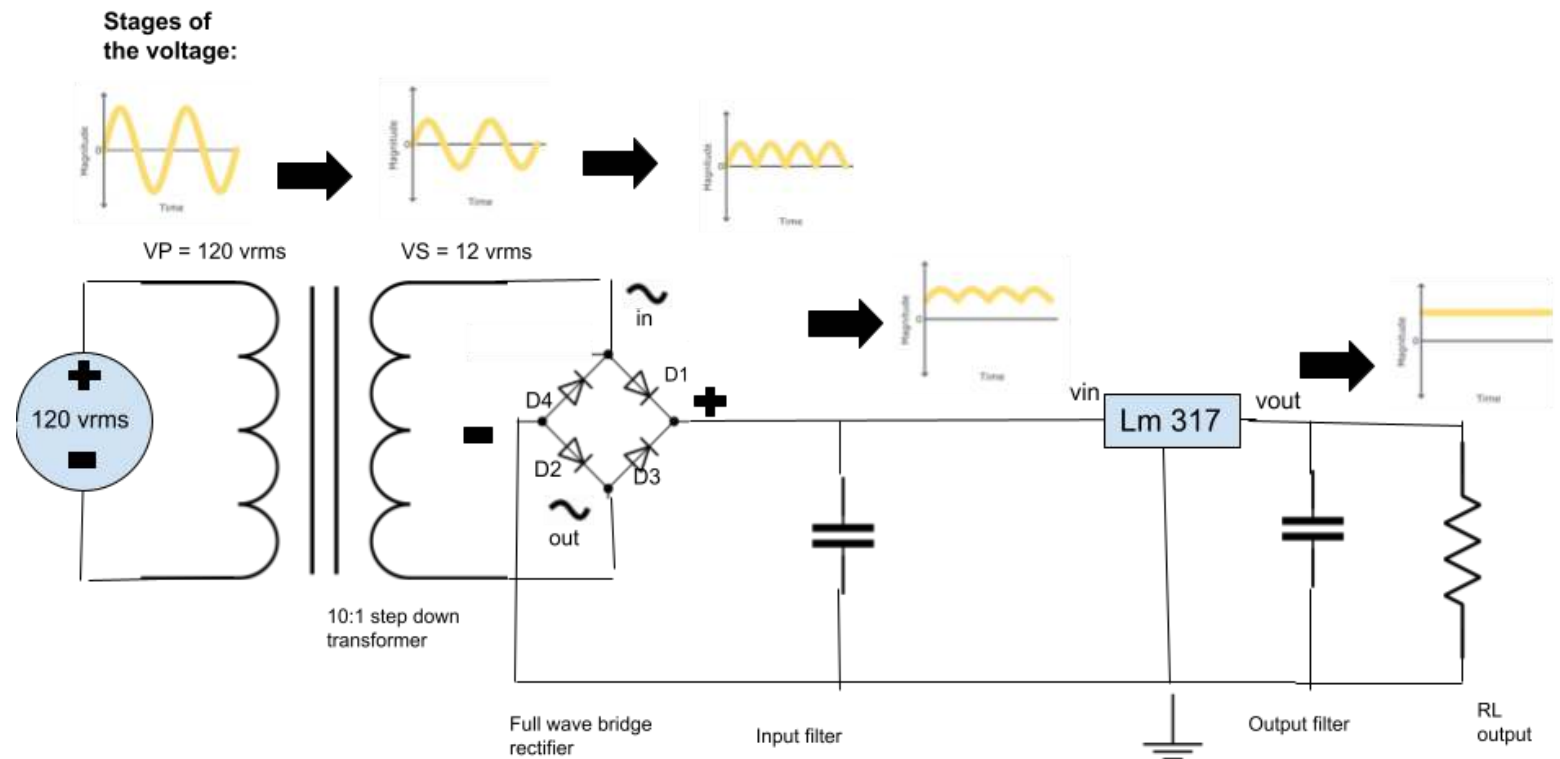
Circuit operation



IMG 1- This is the schematic from the third attempt, using a PCB designed by Alinan US. More information about this PCB can be found in the links section of the report. Note that this PCB is partially incomplete, and the design struggles that came with this are referenced later in the report.

IMG 2- The circuit designed by us (Edwin and Sarah).

On our designed PCB, there are four circuits- the voltage regulator circuit, the oscillator logic circuit, and the logic pen and buzzer circuit.



IMG 3- The stages of the voltage of a power supply, from AC voltage to DC voltage. This information was sourced from in class notes and drawn by hand, except for the graphical images that were found at www.build-electronic-circuits.com/linear-power-supply/. More information about this source can be found in the works cited section of the report.

Image 3 explains how a typical power supply circuit (with an Lm 317) works. Concerning the voltage regulator circuit part of our schematic, it works like a typical power supply circuit as shown in image 3. Our circuit that we built references these concepts learned in class with a few modifications to adjust for a voltmeter and LED lights for status indication. The graphs, with the x axis being time and the y axis being the magnitude of the voltage, show how each part of the circuit influences the output voltage that is the input voltage to the next component.

Essentially, the step down transformer takes in the input AC voltage of 120 vrms from an outlet and has a secondary output voltage of 12 vrms, thus decreasing the magnitude of the voltage. Then the voltage goes through a full bridge rectifier which removes any negative voltage. Note the specific connections of the full wave bridge rectifier (D3, D4, D5 and D6) - a common mistake that people make is not connecting the diodes to the transformer correctly and mixing up where to connect to ground versus the negative of the input AC voltage source. Then the voltage is smoothed out by the input filter (C3 and C4). The input filter filters most of the noise before it reaches the Lm 317, and has a big capacitance to suppress the input ripple voltage. It functions like a bypass filter and smooths the voltage for the Lm 317. Because of the presence of the output filter (capacitor C1) in our circuit, the LM317 can be damaged by capacitive discharge. Essentially, when the capacitor C1 is discharged by the external protection diode D1, the LM317 can be protected by discharging the capacitor C1 through D1. D2 is a protection diode that helps preventing damage to the LM317 during discharge of the filter capacitor C5 (the filter capacitor at the regulating end which has the function of stabilizing and filtering the output voltage)

Then, the voltage goes through the voltage regulator and the magnitude is adjusted according to user input. The output voltage of the Lm 317 goes through the output filter (C1), which smooths out any remaining noise and ensures that the output voltage resembles DC voltage. It has a smaller capacitance compared to the input filter and leads to a DC output voltage across RL, the load. The D7 blue LED is the voltage output operation indication, and R2 is the current limiting resistor for D7. R1 and RP1 work together to form a stable resistance that is also adjustable, and the output voltage can be adjusted by changing the resistance of RP1. Concerning the DC output voltage of our circuit, the LM 317 can measure voltages from 1.25 to 37 volts max. However,

because the transformer has a secondary output of 12 volts, the board can only produce a maximum output voltage of 12 volts, and thus is limited to a range of 1.25 to 12 volts. The minimum output voltage of the circuit is 1.25 due to there being a CD4069 hex inverter on our circuit. Additionally, the voltmeter has a minimum reading of 2.5 volts, so the usable output voltage that can also be read on the voltmeter is between 2.5 and 12 volts. If there is an output voltage, the blue LED (power) turns on.

The oscillator logic circuit (which is essentially a signal generator circuit) uses the IC CD4069UBE provides an output that is a signal for the yellow light (representing the presence of signals). It uses logic gates to generate a repeating square wave to control the switching transistor Q1 that is dependent on the resistor R5- adjusting the resistance value of R5 also adjusts the charge and discharge time, which therefore adjusts the oscillation frequency. IC CD4069UBE has 6 NOT gates and three functions. The first is for the clock oscillator (square waves), the second is for a logic comparator in the status logic circuit (detecting if an input is high or low, which turns on either the red or green lights and activating the buzzer) and the third function is for basic digital signal processing and testing logic on J2 for simulating a logic probe (which was not utilized). Concerning the signal generator circuit, U2A, U2B, R5 and C6 form a non-gate oscillation circuit. The square wave signal is output backwards through U2F, while the square wave status is indicated through the D8 yellow LED. As far as the specifics about how a signal generator circuit works, assuming the U2B output is high, then the U2A input is also high. Likewise, if the U2A output is low as well as the U2B input. If the output at U2B is high, the output pulse of U2B is guaranteed to be high. When the C6 capacitor is charging, the input voltage going to U2A decreases and eventually the U2A outputs a high level and the U2B outputs a low level. Because the U2A outputs a high level, the capacitor is

reverse charged, and then the square wave signal is output backwards through U2F, while the square wave status is indicated through the D8 yellow LED.

The status logic circuit (logic pen circuit) determines which led will turn on- the red led for high voltage or the green led for low voltage. When the input is low, the U2E output is high, and the U2D output is low, and since the U2C output is high, D9 (the green LED) is on. This therefore indicates that there is a low output voltage in the circuit. When the input is high, the U2E output is low, the U2D output is high, and the U2C output is low which means that the red LED (D10) is on, which indicates high voltage.

The buzzer circuit consists of the transistor (Q1) and the buzzer. When there is high voltage (assuming DS03 is connected to the power supply by a wire) of 5V, the buzzer produces a sound. Generally we did not connect DS03 due to convenience and focusing on the required power supply part of the circuit.

Design process

Component list

No.	Component name	Designator	Parameter	Quantity
1	Resistor	R1	240 Ω	1
2	Resistor	R0, R2, R3	1k Ω	3
3	Resistor	R6	10k Ω	1
4	Resistor	R4	100k Ω	1
5	Rectifier diode - 1N4007	D1-D6	DO-41, rating up to 1000V,	6
6	IC socket	U2	DIP-16	1
7	IC- CD4069	U2	DIP-16	1
8	Transistor- S9014	Q1	TO-92	1
9	Blue LED	D7	5mm	1
10	Yellow LED	D8	5mm	1
11	Green LED	D9	5mm	1
12	Red LED	D10	5mm	1
13	Ceramic capacitor	C2, C3	0.1 μ F	2
14	Electrolytic capacitor	C5, C6	10 μ F	2
15	Electrolytic capacitor	C1, C4	680 μ F	2
16	Potentiometer (blue and black)	R5	100k Ω	1
17	Buzzer- LS1	LS1	Active at 5V	1
18	Terminal- KF301-2P	P1		1
19	Terminal- KF301-3P	P2		1
20	Regulator- Lm317	U1		1
21	Heat sink			1
22	Potentiometer (black)	Rp1		1
23	Voltmeter		2.5V- 30V	1

24	Transformer- DB351491	Not mounted	Vp=110v Vs=12v	1
25	Alligator clip			2

Table 1. *Component list for one PCB. Note that there were three attempts, so there were three times the components used in total. Wires, screws, solder materials, the case and the actual PCB is not included in this list.*

For this project, we used a variety of different axial and radial leaded components. Our circuit included many components that were passive (does not introduce energy into the circuit and do not have gains in power) such as the various resistors and capacitors, and many components that were active (usually having a gain and three or more pins) like the S9014 transistor and the integrated circuit (CD4069).

There were many factors to consider when deciding between using through hole or surface mount components. Through hole components are more reliable in harsh environments and better for capacitors and resistors (due to improved heat dissipation). Most relevant to us, they are easier to solder, secure on the PCB and easier to make improvised electrical connections with. This led us to choose through hole components, despite the components taking up more space on the board, being more expensive due to a copper plate also needed to be on the bottom of the board as well as on the top.

There were many factors to consider when choosing capacitors, such as polarization, temperature coefficient (how stable the capacitance is over a range of temperatures), temperature rating (how well it withstands high temperatures, tolerance, etc).

The capacitors on our board are surprisingly small compared to other power supplies. The size of the capacitor is big if either its maximum voltage increases or if its amount of capacitance increases. Since our power supply is limited to a voltage range of 1.25V to 12V, there is not a need for huge capacitors that can withstand a lot of voltage. Therefore having smaller capacitors that meet the required task is reasonable.

Additionally, we used a combination of aluminum and ceramic capacitors. We choose to use aluminum versus ceramic based on the task and also the needed capacitance. Ceramic capacitors are only available for low capacitance values of 10uF and below and are ideal for values less than 1uF, but we used them when we could because they have a high maximum voltage rating (50-100v), small size that fits better on the PCB, were cheaper and had a longer life (although the capacitance does decrease with component age). We used aluminum capacitors when the amount of capacitance needed necessitated this. We had to consider the downsides to using aluminum capacitors (higher cost, less spatially efficient on PCB, a limited life span and needing to be derated for 50%) .

We also had to consider the goal of the capacitors in the circuit, such as how the needs for the input and bypass filter differed based on expected noise levels. The input filter's goal is to direct all the unwanted voltage spikes and noise into the capacitor to protect the lm317, so having an adequate input filter capacitor is critical for the circuit. The input filter needs to act like a bypass capacitor (usually 0.1uF), which acts as a short circuit for high frequency signals (noise), shunting it to ground, and an open circuit to low frequency DC signals. We have a high current and low resistance, which creates a large load and increases ripple voltage thus necessitating using a higher capacitance, because capacitors with a higher capacitance are more effective with filtering

low frequency noise like ripple voltage. But we also need to filter high frequency noise. Therefore we used both a small capacitance of 0.1uF (C3) as well as a large capacitance of 680uF (C4) before the Lm317, to act as the most effective input filter.

For the components, we needed to make sure that the components operated below the maximum voltage and temperature they might receive. This ensures that the circuit is safe and also long-lasting. To find out how much we need to derate for temperature, we can use the formula

$\frac{(\text{operating temperature} - \text{room temperature})}{5 \text{ mW/C}}$, where the room temperature is 25C (which is usually the component's best temperature). Then, at that specific operating temperature the maximum power dissipation (PD) can be calculated as

$$PD_{max} = \text{max power dissipation at room temperature} - \text{amount to derate}$$

To find out how much we need to derate for voltage, we can use the formula $\text{component derating} = \frac{\text{component operating max}}{\text{stress in \%}}$, where stress is $1 - (\text{derating in \%})$. For example, our aluminum capacitors need to be derated at least 50%. This means that the stress is calculated as $1 - 0.5 = 0.5$ and is 50%. Since the circuit gets a maximum secondary voltage from the transformer of 12 volts, the voltage rating for the capacitor needs to be $\frac{12}{0.50} = 24\text{V}$. Realistically, we can get by with a bit of a lower voltage rating due to all the components before the capacitor, so there's some flexibility.

We also had to use this process to ensure that the transformer would operate below the maximum voltage it could handle. Considering the function of the transformer was the most critical thing to consider to ensure that the circuit was safe- it could be dangerous if the transformer was

positioned incorrectly (and was a step up instead of a step down), had electrical connection issues, or received more voltage than it could handle safely. The transformer we found was cheap but was rated for 110v and needed to get an input of 120v. Typically transformers are supposed to only use up to 80% of their rated capacity long term, which means they can be stressed up to 80%. The component derating would therefore be found by $stress = 1 - (derating\ in\ \%)$, or $0.80 = 1 - (0.2)$, meaning that the derating is 20%. Since the component is rated for 110v and we don't want to stress it more than 80% of its maximum rating, the maximum voltage it can technically receive is $\frac{component\ rating}{stress} = \frac{110V}{0.80} = 137.5V$. So a 110v transformer can operate with an input of 120v. Additionally, $\frac{120v}{110v} = 1.09$ or 109%, so 120v is about 9% over its rated voltage. Most appliances can operate safely at +/- 10-20% of their rated voltage, so the transformer should be able to handle anywhere from 100v-130v without problems. Therefore we concluded that this transformer is a safe component to use in the circuit.

Concerning the voltage regulator, we decided to use the Lm317 regulator (a positive adjustable conventional linear regulator, housed in a TO-220 package). Voltage regulators work by sensing the output voltage and then adjusting the current source required by the load to hold the output voltage at the desired value. The output voltage is controlled with a feedback loop. There is a time lag with correcting the output voltage after the load demand changes (called a transient response, how fast the regulator returns to a steady state after a load change) but we assume that the components receiving our power supply's output voltage will not be particularly impacted by this time lag.

The Lm317 regulator has many benefits- it is cheap, easy to find, adds minimal noise into the circuit and most critically, it has a simple design and is easy for us to use and understand. If we did have confusion about using this component, we could have used its datasheet as a reference, but there was no need to. Unfortunately it necessitated a heat sink, which takes up space on the PCB, has only a step down configuration, less efficiency and has a drop out voltage(the minimum difference between the input and output voltage at which the regulator can maintain a stable output) of about 1.5-2v. We decided to use this linear regulator as opposed to a switching voltage regulator- this meant that the transistors inside operate in linear mode as opposed to the pass transistor operating like a switch and being in either saturated or cut off (being fully on or fully off). We made this decision primarily due to the convenience of the Lm317, but this led to a more limited range for the output voltage. We also decided to use a conventional regulator as opposed to LDO. For LDO, the internal pass transistors also operate in linear and active mode. LDO regulators have a low dropout voltage ($V_{in}-V_{out}$ is smaller than conventional linear regulators and is approximately 0.5V) which is definitely more ideal, but using LDO would complicate the circuit, be an added expense and result in a lower output current.

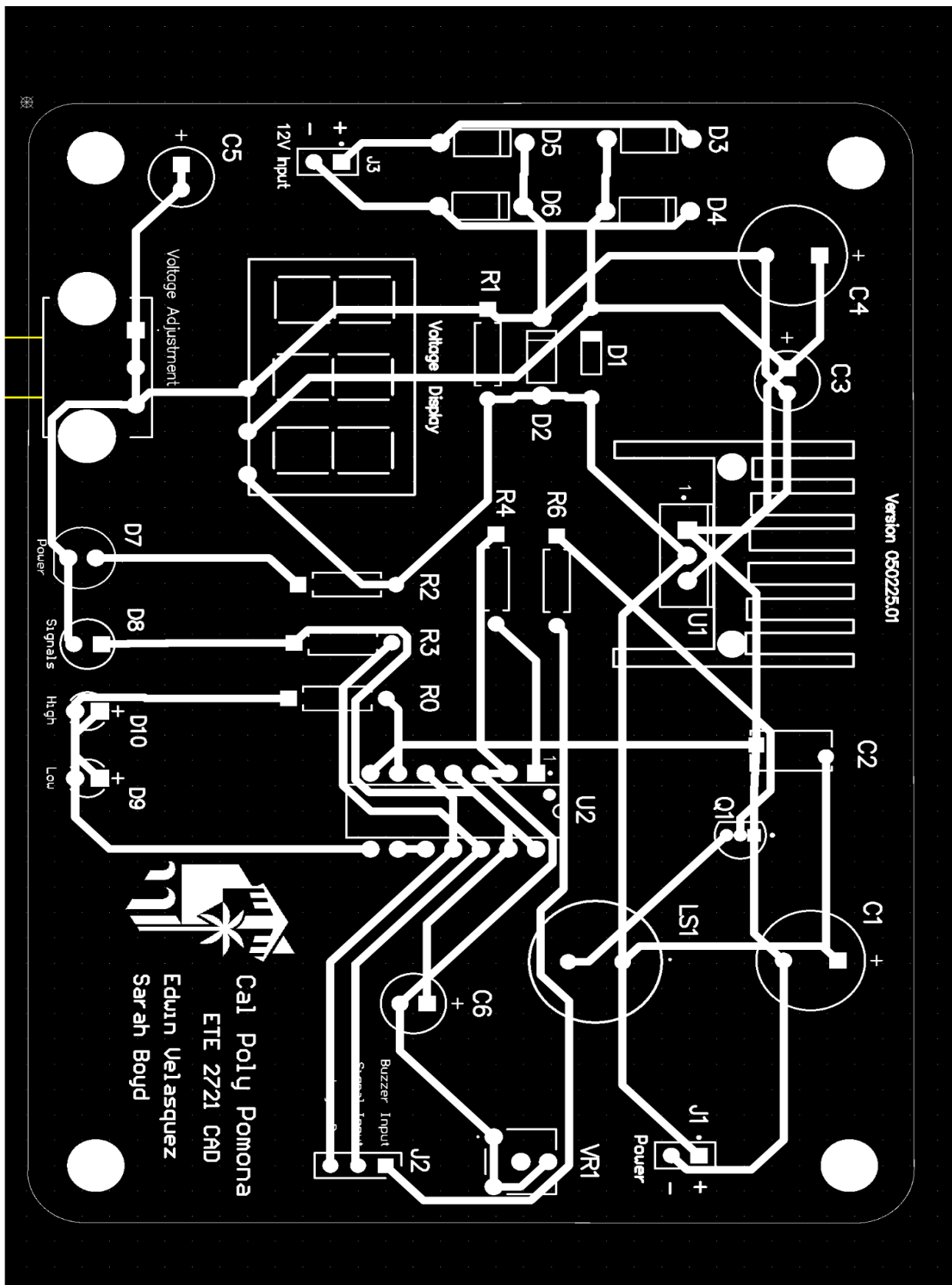
Design struggles

We encountered many struggles while designing our PCB. We referenced the pre-designed PCB sourced from China by Alinan US, and one significant challenge we faced was that much of the documentation was incomplete or only available in Mandarin. This particularly impacted the design process for the schematics of the circuit. The provided schematic lacked essential components, particularly the voltmeter, which is used to display voltage via three 7-segment displays. These displays are controlled by a driver IC that is regulated through a potentiometer. To

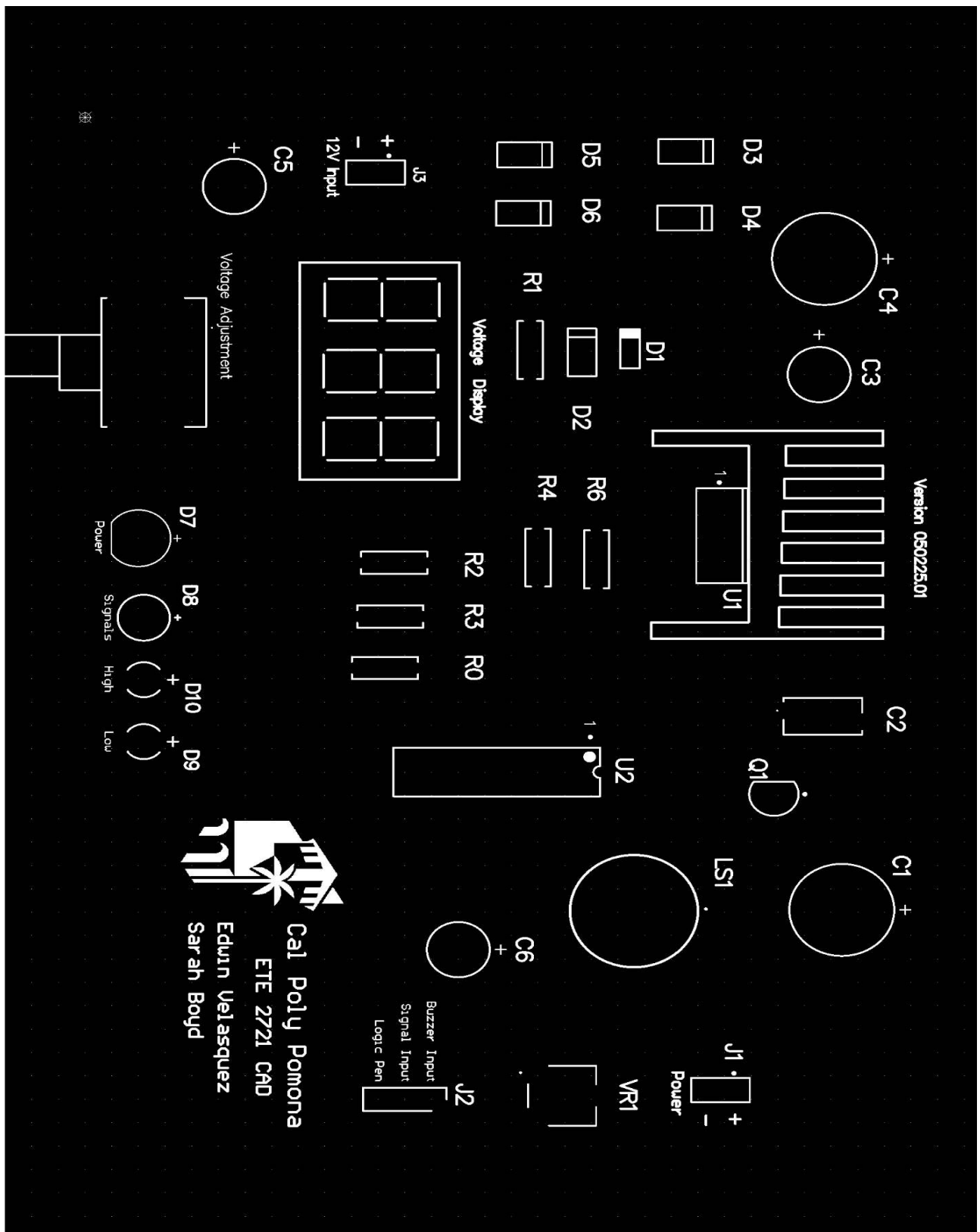
address this issue, we used a digital multimeter to trace the signal path from each of the three connection pins on the voltmeter. We then replicated this tracing on our PCB to successfully integrate the missing component.

Using Altium

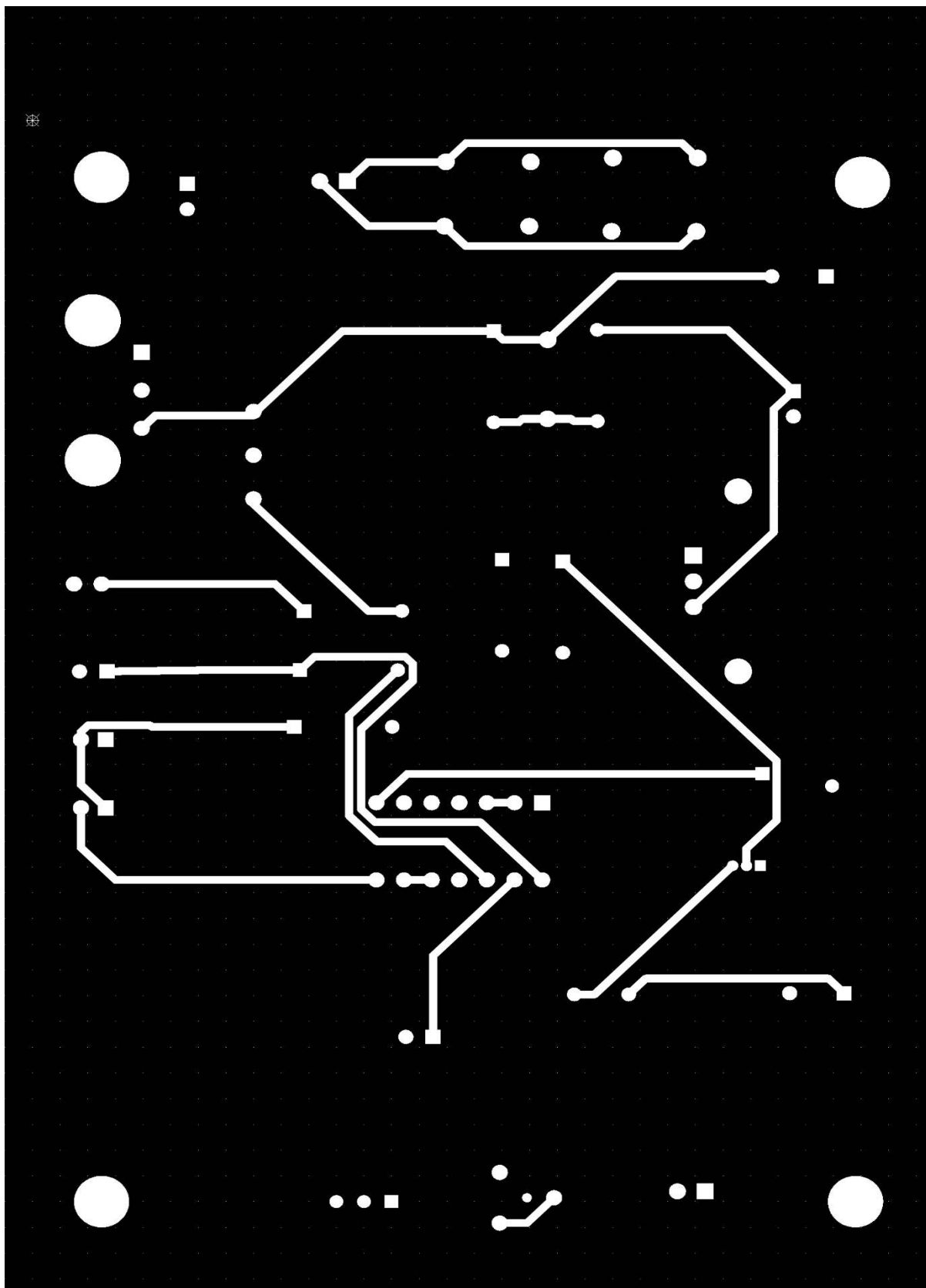
In the following pages, the schematic is exported in black and white, along with the top copper, bottom copper and silk screen.



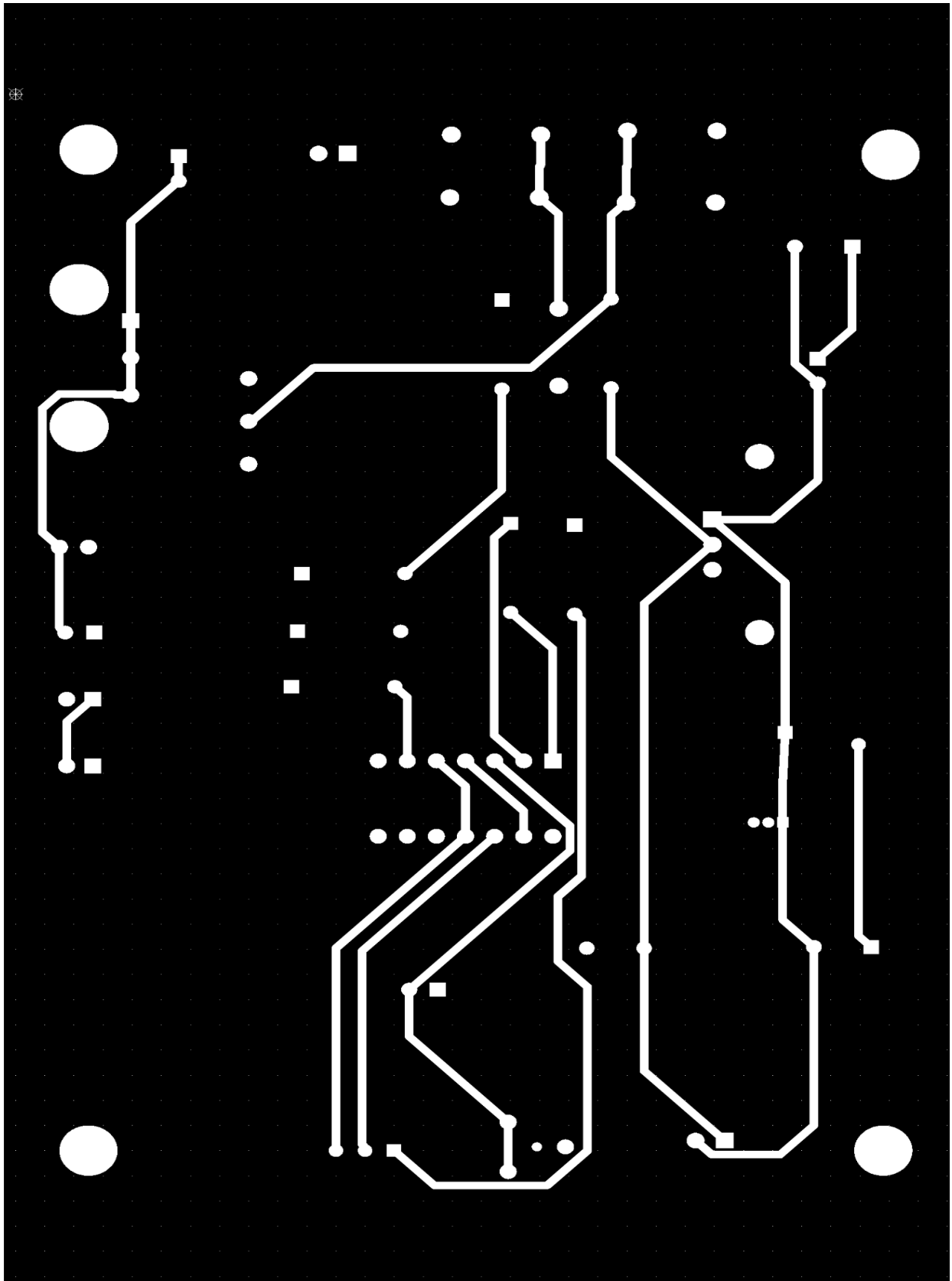
IMG 4- schematic in black and white



IMG 5- silkscreen in black and white



IMG 6- *top copper in black and white*



IMG 7- *bottom copper in black and white*

While designing our board, we needed to not just keep in mind our goals for the circuit but also keep in mind the capabilities of the PCB manufacturer (designing for manufacturability). Just because a PCB can be designed in Altium does not automatically mean that our manufacturer (JLC) can actually produce it. This is why a DFM check (which is done after we submitted the Gerber files to JLC) is the first step of the PCB manufacturing process to ensure that no rules (such as for thickness, clearance) are violated. As designers we had to prioritize minimizing potential issues. Re-ordering our PCB was out of the question due to time constraints and financial constraints (it cost approximately \$90 per order). We were thoughtful with aspects such as minimum trace width, component placement, clearances, tolerances, heat management and fixing design rule violations.

The following is a list of design regulations and aspects we kept in mind when designing our PCB:

- The PCB must be double-layer at least to optimize signal routing and power distribution.
- We need to maintain proper spacing between traces (minimum 6 mils) to avoid short circuits.
- Signal traces should be kept as short as possible to minimize noise and interference. Additionally, we need to avoid “criss-crossing” connections.
- Power traces must be sufficiently wide based on the current load, with a minimum width of 20 mils for signals and a greater width for power.
- Components should be logically placed- related components of each sub circuit should be placed in groups on the PCB. For example, components related to the power supply should be close in proximity and not spread throughout the PCB. DOing this helps keep traces short, which reduces parasitic effects.

- Components should be placed keeping its physical dimensions and orientation in mind, mind- this is crucial for when we actually assemble and solder the circuit.

- Ground planes should be utilized to enhance signal return paths and minimize interference. It is recommended to use a ground layer to ensure all components share the same reference point. We should avoid breaking up the ground planes into smaller pieces, and can use vias to do so.

- Minimize the use of vias to avoid added resistance unless necessary.

- All pins must be clearly and correctly labeled in the schematic. This is important for components that have polarity, such as capacitors, or chips where pin 1 serves as the reference for the other pin order.

In Altium Designer, we either created or verified footprints and symbols to match the real-world components that were pre-selected based on the needs of the circuit. For instance, the footprint for the IC was pre-designed by Altium according to its datasheet to ensure that the pin layout and dimensions were accurate. For parts such as resistors and capacitors, we opted for standard package sizes that matched our physical components, specifically 0805 and 0603. Additionally, we placed connectors and headers strategically to facilitate easy assembly. Other aspects, such as the shape of the board, the positions of the holes, and parts not provided by Altium, needed to be designed by hand. The board size is determined by the amount of space required by the components. The hole placement can be determined by deciding the board size and then by doing math, or by simply putting them wherever visually looks right. Parts like the heatsink and buzzer had to be designed by hand using measurements from their data sheets and sometimes estimates.

Concerning designing the traces, for higher voltages we needed to design larger spacing between traces to avoid arcing. For large currents, we use wide traces to reduce heating and voltage drop across the trace. For small currents traces can be narrower with little change ($V = IR$) - if the current is small then the voltage drop across the trace will be small too, regardless of trace resistance, and voltage arcing cannot occur. Most manufacturers have a capability of 5-6 mils for width and spacing, but a small trace spacing (the distance between adjacent traces on a given layer) makes it difficult for the manufacturer to make the PCB without errors (such as unwanted shorts or opens). Due to this, we set it to 10 mils for ease of design.

Additionally, a high current going through the trace can cause a voltage drop, the trace to overheat and even burn up and disappear! Increasing the trace width increases the area of the trace and thus decreases the resistance and is better for higher voltage. Recall that the resistance of a trace is determined by the length and also area of a trace, based on the formula resistance = $\rho \frac{Length}{Area}$. Because the resistance is also determined by the length of the trace, we want to make the trace as short as possible. The length and width of the trace is changeable, but the height is considered to be a constant due to it being harder to control. The standard height of a trace is 0.0014 inches (or 1.4 mils). High frequency traces basically act like antennas, which can radiate or pick up noise and EMI, which reduces the circuit performance and can interfere with nearby devices. If we design a lot of long traces in our circuit, this could impact the performance and any circuits or components connected to the output of our power supply.

We also needed to design our PCB with ground and power planes in mind. Ground and power planes are large areas of conductive metal that all IC ground and power pins should go to, and are essentially a very large trace. These planes provide low impedance and shielding from EMI

and reduce noise, and are especially good for sensitive and high frequency circuits. Including a ground and power plane was not especially critical for our project due to all our filter capacitors and because our traces were already short and wide, but for future redesigns this would be good to include. For PCBs including ground and power planes, we as designers need to avoid breaking these planes up by using vias to jump to a different layer and prevent trace paths from going through planes.

In our particular project, our PCB schematic was simple enough to use two sheets of copper and not necessitate many layers. But for other future designs, it is important to consider the typical function of each layer of the PCB. Typically, the top layer is for high speed signal traces, important traces, and control. The inner layer (second from the top) is usually a ground plane with the added bonus of shielding the layers (minimizing cross talk between layers). Vias are used to connect things to this layer. The next inner layer (third from the top) is typically the power plane. Sometimes it is divided in multiple sections if there is a need for different voltages in the circuit, but for simpler circuits only one power plane for the whole board is needed. The bottom layer is usually another signal layer for less critical or simpler traces.

Gerber file

We used Altium Designer to generate the Gerber files after completing the PCB layout. First, we conducted a Design Rule Check (DRC) to ensure there were no errors. Next, we utilized the "Fabrication Outputs" feature to export the essential Gerber layers, including the Top and Bottom Copper, Solder Mask, Silkscreen, and the NC Drill file. All files were saved in a ZIP folder and prepared for PCB manufacturing through [JLCPC.com](https://www.jlcpcb.com), a company located in the People's Republic of China.

Status Report - Notepad

File Edit Format View Help

Output: NC Drill Files

Type : NC Drill

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Generated File[PCB1.TXT]

Generated File[PCB1.LDP]

Generated File[PCB1.DRR]

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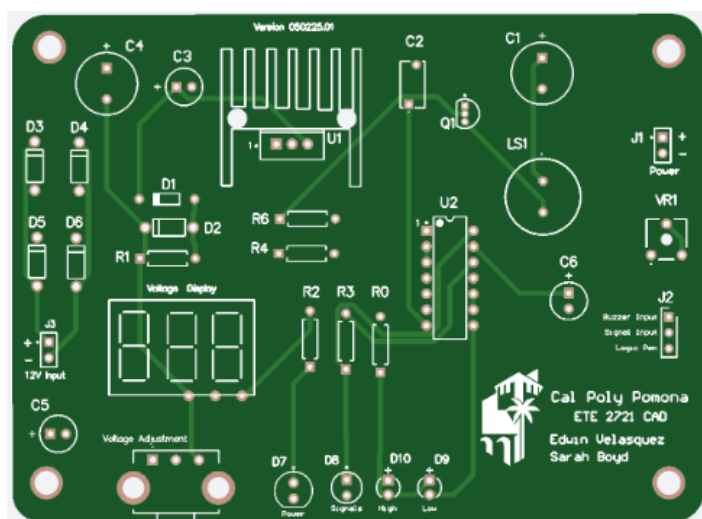
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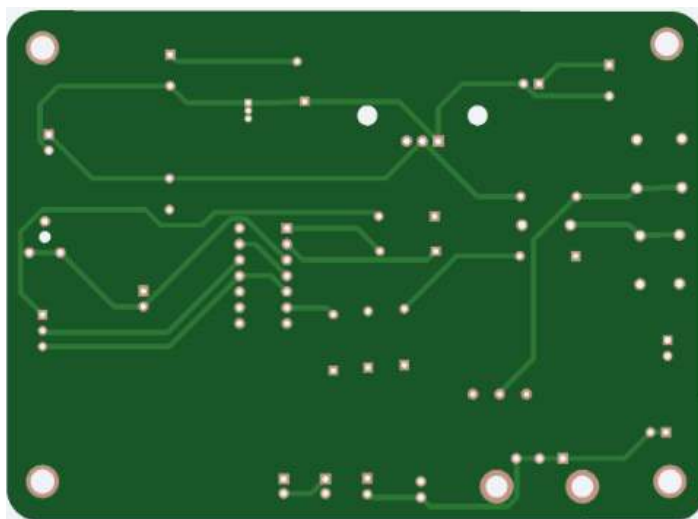
Status report of PCB

Name	Type
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PCB1	Report File
PCB1.EXTREP	EXTREP File
PCB1_Copper_Signal_Bot	CAMtastic Layer Gerber Data
PCB1_Copper_Signal_Top	CAMtastic Layer Gerber Data
PCB1_Legend_Bot	CAMtastic Layer Gerber Data
PCB1_Legend_Top	CAMtastic Layer Gerber Data
PCB1_NPTH_Drill	CAMtastic Layer Gerber Data
PCB1_Profile	CAMtastic Layer Gerber Data
PCB1_PTH_Drill	CAMtastic Layer Gerber Data
PCB1_Soldermask_Bot	CAMtastic Layer Gerber Data
PCB1_Soldermask_Top	CAMtastic Layer Gerber Data
PCB1-macro.APR_LIB	APR_LIB File

IMG 9. Contents of the Gerber file



IMG 10. PCB as generated by JLC



IMG 11. PCB as generated by JLC

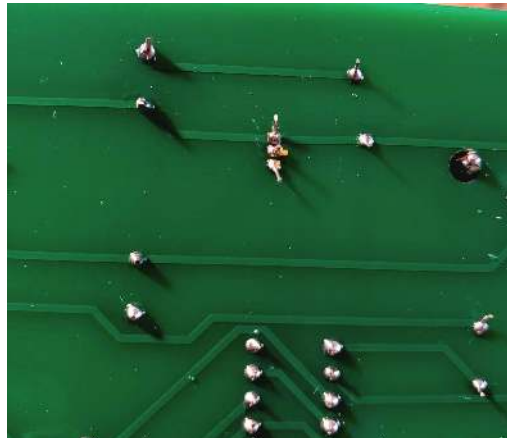
Soldering Process and Project Results

For soldering, we used tin-lead (SnPb), with 63% tin (Sn) and 37% lead (Pb). We used tin lead rosin core solder (0.6mm, 100g, 1.8% flux) from the brand MAIYUM. We turned our soldering iron to 370C, and connected our components with the solder forming a volcano shape as opposed to a sphere. The rosin core solder we used has flux inside the wire itself, which is important because flux is important and used to improve the quality of the solder joint, cleans surfaces of materials being soldered, removes oxidation and helps the solder to adhere better.

While soldering, we did encounter a few issues. The potentiometer to adjust the voltage was oriented incorrectly on our PCB, so we had to strip the insulation from some wires and connect the potentiometer to the PCB through wires. The voltmeter display had a similar issue. Both components were designed by hand on Altium, so the orientation errors were due to our mistakes. There were also some components that did not fit on the board due to the pin spacing being slightly off, and we went through a similar process. Additionally, the spacing for the transistor (Q1) component didn't technically violate the design rules, but it was very difficult to solder by hand. Because Sarah's soldering iron tip was too big, it was very hard to prevent the soldering iron from forming a short circuit with nearby components. Edwin had the proper soldering iron tips and was able to make a good electrical connection with the solder. Professor Mandala very kindly helped us solder this connection. In the future, we need to check that the components with the data provided by Altium are still workable for hand soldering, even if they don't technically violate design rules.



IMG 11. Q1 connection, with nail for scale



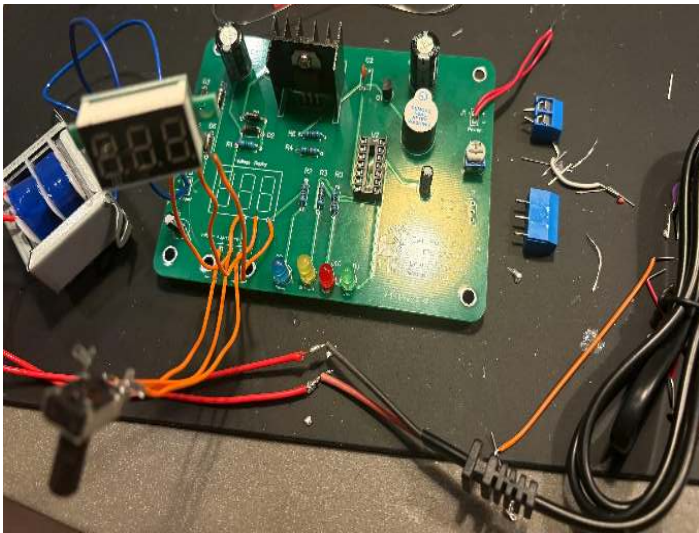
IMG 12. Soldered Q1 connection, after some tries



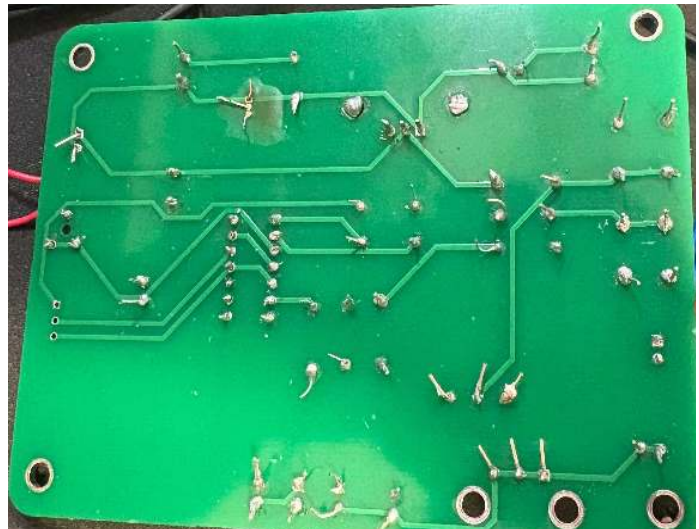
IMG 13. Soldering station

Results of the three attempts:

The first attempt was soldered by Sarah. The circuit only gave an output of negative voltage. This issue was not caused by the orientation of the diodes on the full wave bridge rectifier as they are oriented correctly on the schematic, PCB layout and on the physical board. The orientation of the LM317 also seems correct.

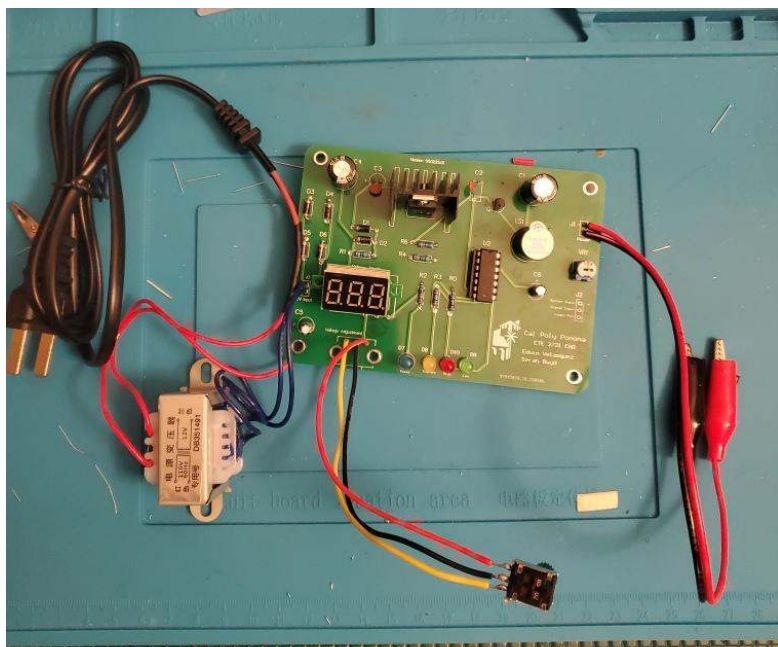


IMG 14. Attempt 1, soldered and assembled



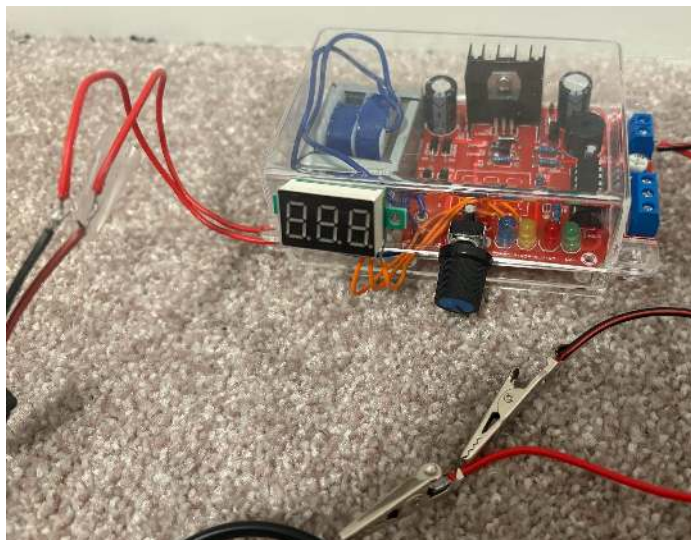
IMG 14. Soldered electrical connections

The second attempt was soldered by Edwin. Both PCBs output a small amount of negative DC voltage, indicating that there was a design or component error rather than a soldering mistake.

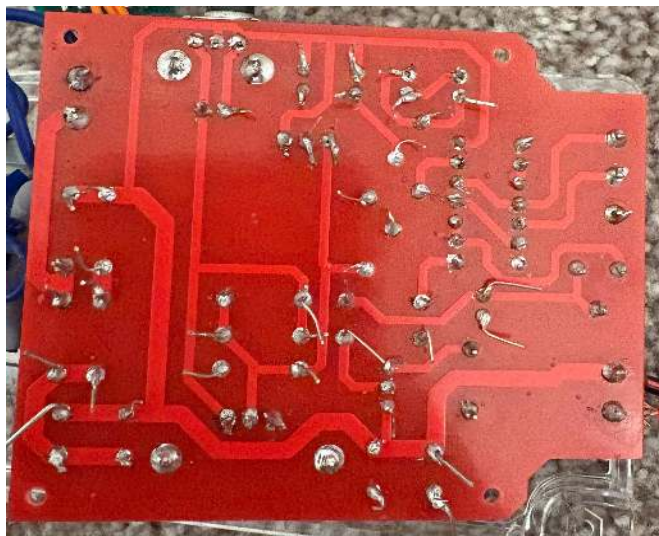


IMG 14. Attempt 2

The third attempt, using the PCB from Alinan US, was soldered by Sarah, and it was successful!



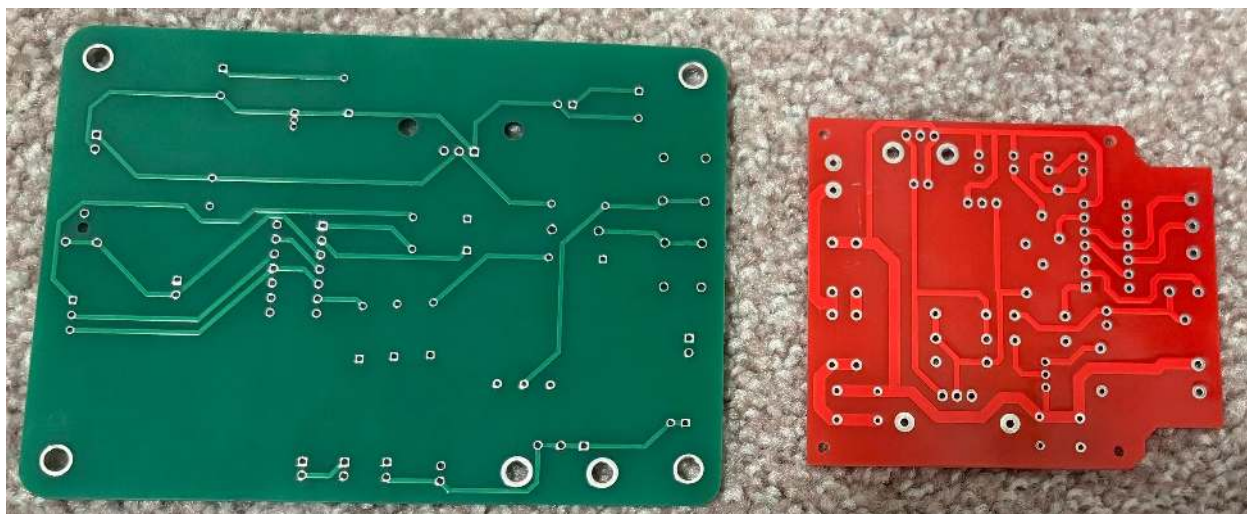
IMG 16. Attempt 3



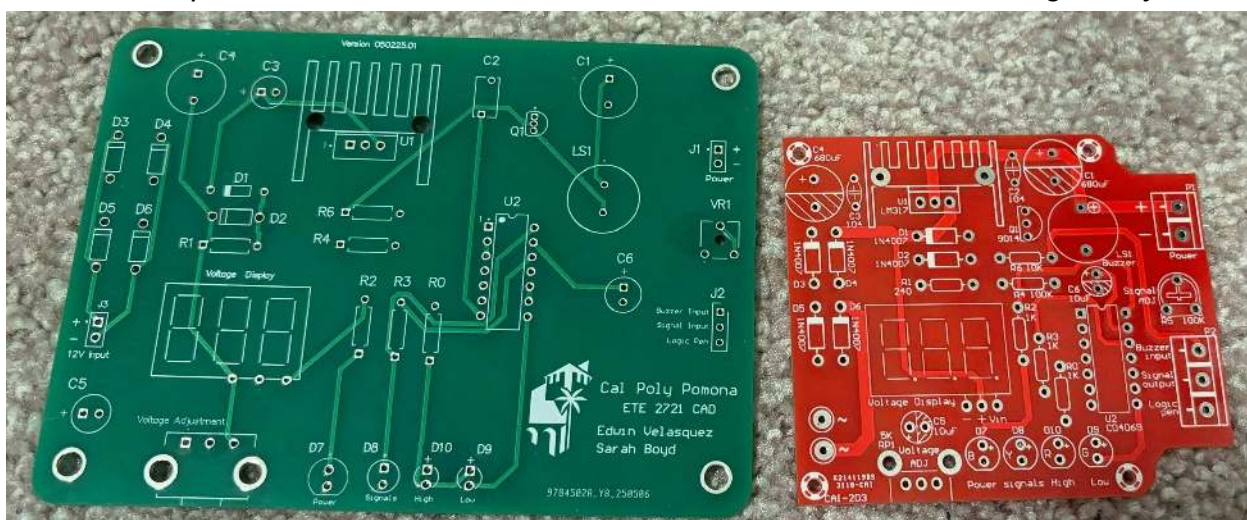
IMG 17. Attempt 3

The working project can be found using this link:

<https://drive.google.com/file/d/1vBi0UU9jklbOX2lYrYp1fBSshVTw9b37/view?usp=sharing>



IMG 18. Comparison between the PCB from Alinan US and the PCB designed by us.



IMG 19. Comparison between the PCB from Alinan US and the PCB designed by us.

Enclosure

There were a total of three different enclosures utilized for this project. The first enclosure was premade by Alinan US. This enclosure can be seen in image 16.

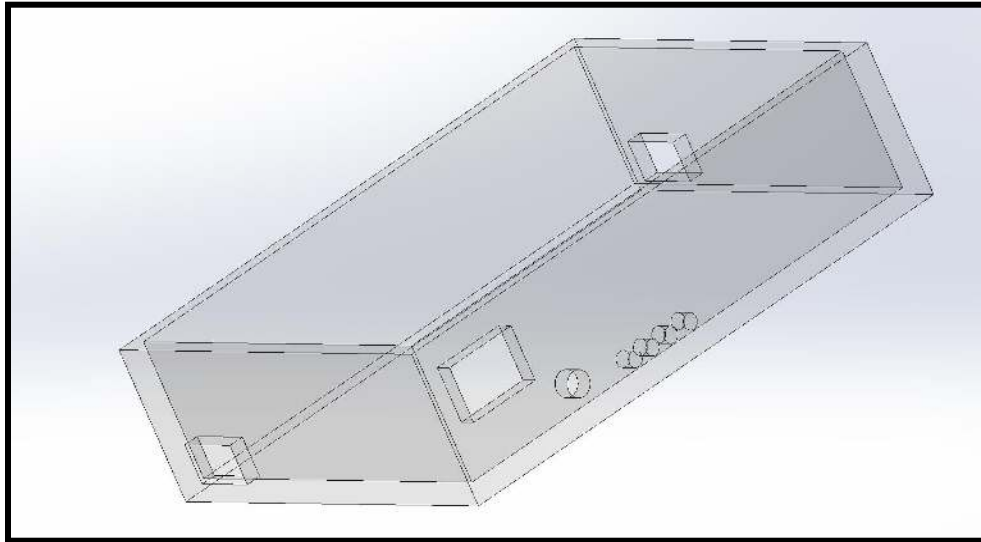
The second enclosure was purchased with the goal of drilling holes to fit the voltmeter, potentiometer, three LEDs and wires for the input AC voltage and output DC voltage. There were issues creating the holes in the enclosure due to issues with the tools needed to do so. This enclosure was still used for the project presentation.



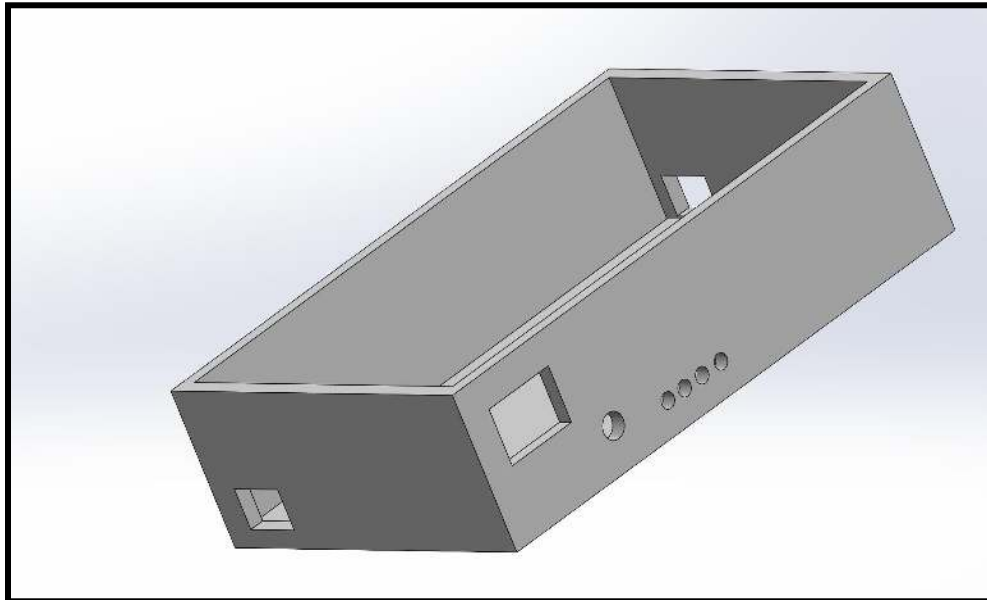
IMG 18. *Enclosure used for the presentation.*

The third enclosure was designed using SolidWorks and intended to be printed with clear plastic. Unfortunately the clear filament was defective, so a red one was used instead. It

took a longer time than expected to print (over 10 hours) , and there were issues with the dimensions of the 3D printer because the maximum dimensions of the printer is 15mm x 15mm x 15mm. Due to the set backs, this enclosure was not created until after the presentation.



IMG 19. *Enclosure made with clear plastic on SolidWorks*



IMG 20. *Enclosure made of an alternate material*



IMG 21. Enclosure



IMG 22. Enclosure



IMG 23. Enclosure

Conclusion

This report documents the stages and struggles of designing our power supply. Before starting work on the PCB, we thoroughly analyzed the entire circuit to understand the signal flow and how components interact with each other. We checked each component's footprint and orientation to prevent placement errors. Due to missing documentation, we reverse-engineered part of the circuit using a multimeter and logical tracing techniques. We also utilized Altium Designer's Electrical Rule Check (ERC) to identify common errors, such as unconnected pins. Throughout the process, we continuously tested for continuity and monitored reference voltages to ensure proper functionality. We had many various struggles with each stage of this process, which were previously mentioned in sections covering the design process using Altium and the soldering process. The power supply designed by us (attempts one and two) did not successfully output the correct DC voltage, but it did output some voltage. Attempt three, which was the PCB designed by Alinan US, was supposed to be a backup and ended up being the only attempt of this project that was truly successful. Our hypothesis about why our PCB might not have worked was previously covered in this report- after ruling out a potential mis-orientation of the diodes in the full wave bridge rectifier, this leads us to think that there was an orientation issue with the Lm317 regulator.

Despite so many struggles and unsuccessful attempts, we learned a lot from this project. We learned how to troubleshoot unclear schematics and documentation, determine what components to use and how to determine if the components are safe, how to design a PCB on Altium and DFM,

how to get Gerber files, how to solder, and much more. When designing boards on Altium in the future, we will keep the following in mind:

- Always double-check the footprints and pinouts of components using their datasheets before placement. This avoids designing a component in opposite orientation or reverse polarity.
- Utilize net labels and consistent naming conventions to make schematics easier to understand and troubleshoot.
- Frequently run Electrical Rule Checks (ERC) and Design Rule Checks (DRC) during the design process to catch errors early.
- Organize components logically to simplify routing and minimize trace lengths.
- Use polygon pours for ground planes to enhance signal integrity.
- Keep track of design changes and document them for easier debugging later on.

Links

Links regarding Alinan US PCB:

- <https://www.google.com/url?q=https://www.amazon.com/Adjustable-Regulator-PEMENOL-Electronic-Multifunctional/dp/B07SDLRSGD%23averageCustomerReviewsAnchor&sa=D&source=docs&ust=1747523654405947&usg=AOvVaw0j4Tdo896IGH-h-HuakBQw>
- <https://www.google.com/url?q=https://www.google.com/url?q%3Dhttp://www.diyleyuan.com/JC/X203.html%26sa%3DD%26source%3Ddocs%26ust%3D1747519596689803%26usg%3DAOvVaw0bKrPL91kprSiHbws5G4Wg&sa=D&source=docs&ust=1747525913847518&usg=AOvVaw1E2O5W2yxZu4Vi1qyY4nvr>

Link to working PCB project:

- <https://drive.google.com/file/d/1vBi0UU9jklbOX2lYrYp1fBSshVTw9b37/view?usp=sharing>

Works cited

- Urias, Omar Muñoz. "How to Build a DC Linear Power Supply." *Build Electronic Circuits*, 1 Sept. 2023, www.build-electronic-circuits.com/linear-power-supply/.
- Utsource, Utsource, and Instructables. "DIY Power Supply Using LM317: LM 317 Variable Voltage Output." *Instructables*, Instructables, 25 Jan. 2020, www.instructables.com/DIY-Power-Supply-Using-LM317-Lm-317-Variable-Volta/.

The first source, "How to Build a DC Linear Power Supply.", was used in the circuit operation section for images of graphs of a power supply.

The second source, "DIY Power Supply Using LM317: LM 317 Variable Voltage Output", was used as a reference for using a voltmeter in a power supply circuit, and how that impacts the circuit functioning and schematics.

Bill of Materials (BOM)

A1	Line #														
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
	Line #	Name	Description	Designator	Revision ID	Revision State	Revision Status	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacture Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
1		Volmeter V2	Idea tab		CHP-019-00046-5	Draft	Up to date	1							
2		ECA-1E1102	Aluminum Electrolytic C1, C4		CHP-001-00023-8	Draft	Up to date	2	Panasonic	ECA-1E1102	Volume Production	Amnet	ECA-1E1102	0.15792	0.31584
3		S101C25SNNK16R	VISHAY - S101C25C2		CHP-001-00086-1	Draft	Up to date	1							
4		CAP 01uF 2.5mm	Aluminum Electrolytic C3, C5, C6		CHP-001-00090-2	Draft	Up to date	3	Panasonic	ECA-1E1102	Volume Production	Amnet	ECA-1E1102	0.15792	0.47376
5		1N4148	High Speed Diode, D1		CHP-004-00000-2	Draft	Up to date	1	ON Semiconductor	1N4148	Volume Production	Amnet	1N4148	0.00755	0.00755
6		1N4007	General Purpose Rectifier D3, D4, D5, D6		CHP-004-00007-2	Draft	Up to date	5	Diodes	1N4007-1	Obsolete	Farnell	1543998	0.27194	1.36
7		151051BS34000	Single Color LED B107		CHP-007-00036-1	Draft	Up to date	1	Wurth Electronics	151051BS34000	Volume Production	Digley	732-5015-ND	0.24	0.24
8		LT1-1C1HE	LED Vln-Color Red B8		CHP-007-00020-2	Draft	Up to date	1	Vishay Lite-On	LT1-1C1HE	Volume Production	Digley	189-1709-ND	0.19	0.19
9		304090040	Green LED, 80 mW, D9		CHP-007-00006-2	Draft	Up to date	1							
10		304090041	Red LED, 80 mW, 3/D10		CHP-007-00007-2	Draft	Up to date	1	Thiad Magnetics	F-13K	Volume Production	Newark	51V4828	5.03	5.03
11		F-13K	Power Transformer D100		CHP-13501-000105-1	New From Design	Up to date	2	Hawthorn	MD0-9990245	Volume Production	Digley	952-2261-ND	0.12	0.24
12		MD0-9990245	CONN HEADER VEH1, J3		CHP-002-00077-1	Draft	Up to date	1							
13		MD0-9990245	2.54mm (0.1") Pitch J2		CHP-009-00000-2	Draft	Up to date	1							
14		PS124P02B1	Piezoelectric buzzer LS1		CHP-013-00035-1	Draft	Up to date	1							
15		2N3904	Small Signal Bipolar Q1		CHP-009-00005-4	Draft	Up to date	6	Vicoro	MFR-25F8F52-4K75	Volume Production	Amnet	MFR-25F8F52-4K75	0.00756	0.04536
16		MFR-25F8F52-4K75	Adjustable Output, U1		CHP-019-00023-3	Draft	Up to date	1	ON Semiconductor	LM4317G	Volume Production	Newark	98H0572	0.455	0.455
17		LM4317G	Inverter, 4000V/40V U2		CHP-009-00184-1	Draft	Up to date	1	Texas Instruments	CD4690BE	Volume Production	Arrow Electronics	CD4690BE	0.7563	0.7563
18		CD4690BE	Variable Resistor VRF1		CHP-009-00184-1	Draft	Up to date	1							
19		601030	Variable Resistor VRF2		CHP-009-00184-2	Draft	Up to date	1							
20		PT20K-F20BR5K	Variable Resistor VRF2		CHP-009-00184-2	Draft	Up to date	1							
21															
22															